ICU Mortality Prediction Using Machine Learning: A Practical Framework for Risk Stratification

Course : Data Science CS3072

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HEALTHCARE AI

MACHINE LEARNING

Introduction

ICU patients are critically ill and may deteriorate rapidly, requiring continuous monitoring and timely intervention.
Traditional scoring systems such as APACHE and SOFA are widely used, but they may not fully capture complex relationships between patient conditions, treatments, and outcomes.

Recent advances in machine learning allow models to learn patterns from clinical data and analyze multiple predictors simultaneously.
This enables more flexible and data-driven mortality risk estimation to support early detection and clinical decision-making.

Predict Risk

Estimate mortality risk using multiple clinical features and patient conditions.

Support Decisions

Assist clinicians in prioritizing care and allocating ICU resources effectively.

Early Detection

Enable early identification of high-risk patients for timely intervention.

(Esteva et al., 2019)

Problem Definition

The goal of this project is to estimate the mortality risk of ICU patients using structured clinical data. We aim to **identify high-risk patients** and understand which clinical features contribute most to increased risk.

Class 0 — Survival

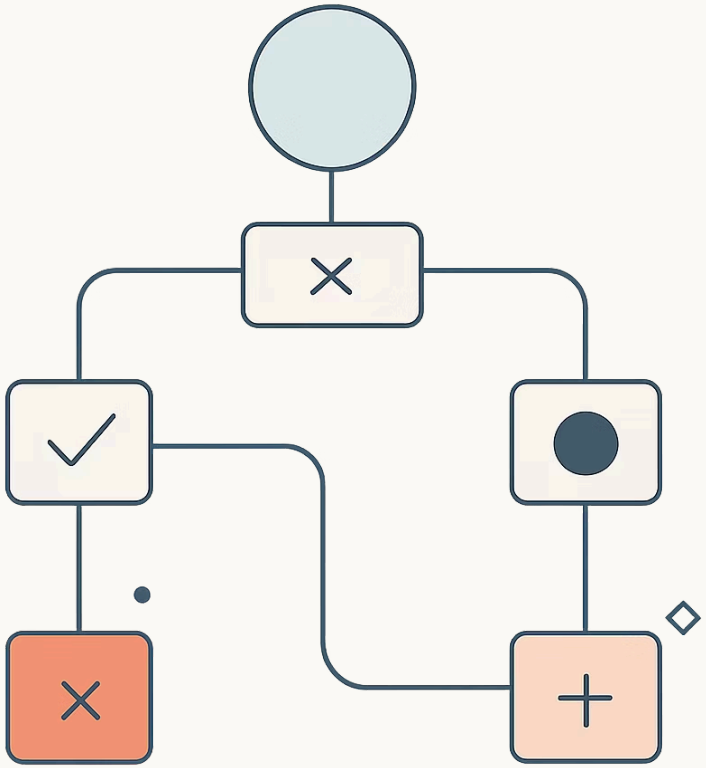
The patient survives the observation period and represents the majority class.

Class 1 — Death

The patient does not survive and represents the critical minority class.

Instead of only predicting survival or death, the model estimates a **risk probability** to support early intervention and patient prioritization.

Since missing high-risk patients is critical, the model is designed to **maximize recall** for Class 1.



(Sendak et al., 2020)

State of the Art & Research Gap

Machine learning models are widely used for ICU mortality prediction, but several limitations reduce their effectiveness in real-world clinical settings.

Limited Generalizability

Many models are trained on specific patient groups or single-hospital datasets, which limits their applicability to broader ICU populations.

Data Quality & Complexity

Clinical data is often noisy, incomplete, and inconsistent, which reduces model reliability and complicates real-world deployment.

Imbalance & Misleading Performance

ICU datasets are highly imbalanced, where survival cases dominate. Models may achieve high accuracy but still fail to detect high-risk patients.

Our Contribution

This project addresses these gaps by using a small set of clinically meaningful features, handling class imbalance, comparing multiple models, and applying threshold tuning to improve detection of high-risk patients.

(Kim et al., 2019; Li et al., 2021; Zhang et al., 2020)

Project Significance, Sustainable Development Goals & Vision 2030

SDG 3: Good Health and Well-being

This project supports SDG 3 by applying machine learning to predict patient mortality risk and enable early clinical intervention.

SDG 9: Industry, Innovation, and Infrastructure

This work contributes to SDG 9 by using advanced data-driven technologies to improve healthcare systems and decision-making.

Saudi Vision 2030

This project aligns with Saudi Vision 2030 by promoting digital transformation in healthcare and supporting data-driven clinical solutions.

Project Significants

This project focuses on recall to detect high-risk patients, uses 7 key clinical features, and applies balancing and threshold tuning to improve real-world reliability.

Dataset Overview

Dataset Composition

The dataset contains **15,000 ICU patient records** used to predict mortality risk.

It includes **clinical and physiological features** collected at or near admission, describing the patient's health condition.

- Vital signs and laboratory measurements
- Patient characteristics (e.g., age, medical conditions)
- Clinical indicators related to patient status
- Target variable: **mortality_label** (0 = survived, 1 = died)

(Kaggle ICU Dataset)

Key Characteristics

Attribute	Detail
Task Type	Binary Classification
Target Variable	mortality_label
Number of Samples	15,000
Number of Features	24
Feature Types	Numerical & Categorical
Class Distribution	Imbalanced (fewer mortality cases)

Data Processing

The dataset was preprocessed to improve data quality and ensure reliable model performance.

Remove Non-Informative Features

Dropped identifiers and irrelevant columns that add noise without predictive value.

Handle Missing Values

Imputed or removed incomplete records to ensure consistent input across all features.

Encode Categorical Variables

Converted categorical fields into numerical format for model compatibility.

Scale Numerical Features

Applied standardization to ensure consistency across models and prevent feature dominance.

Train/Test Split

Divided the dataset into training and testing sets using a stratified split to preserve class balance.

Raw Data

Unprocessed clinical records with noise and **missing values**

Processing

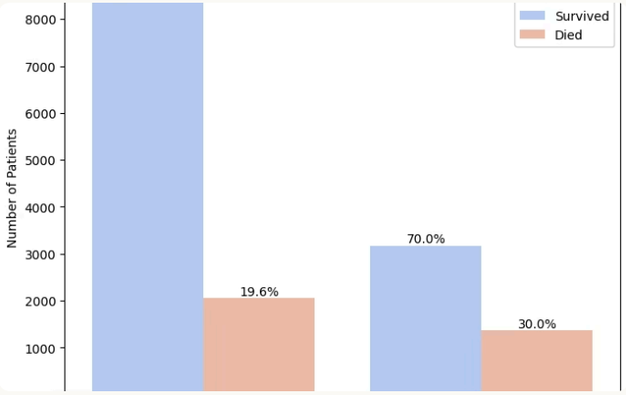
Cleaning, **encoding**, **scaling**, and splitting

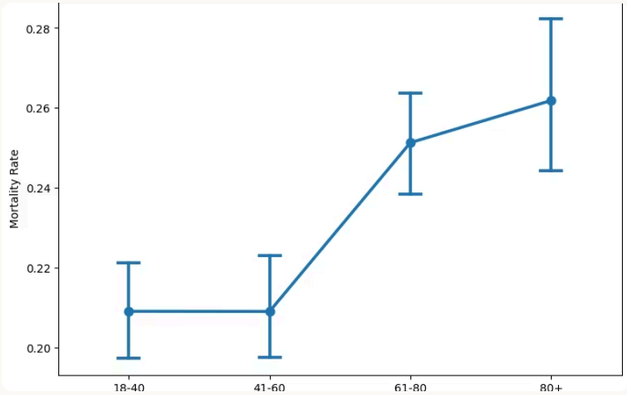
Model-Ready Data

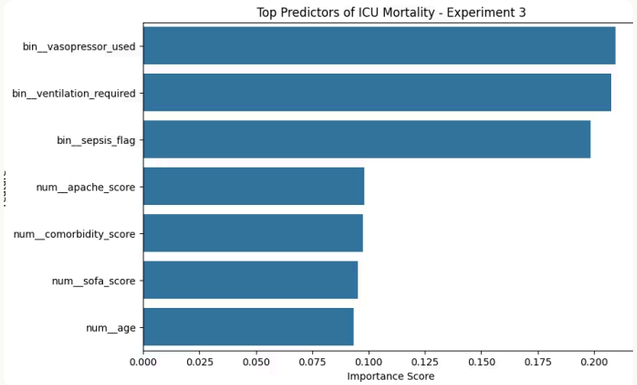
Clean, structured, and balanced dataset ready for training

Exploratory Data Analysis (EDA)

EDA was conducted to uncover distributional patterns, variable relationships, and potential data quality issues before modeling. Key visualizations guided feature engineering and selection decisions.







Ventilation vs Mortality (Bar Chart)

What

Comparison of mortality outcomes based on ventilation requirement.

Why

To understand how critical interventions relate to patient survival.

How

Grouped bar chart comparing survived vs died for ventilated and non-ventilated patients.

What we learned

Patients requiring ventilation have a significantly higher mortality rate, indicating severe clinical condition.

Note: There exists multiple other diagrams that showcases the relationship between the feature(s) and the mortality label that are similar to this one therefor we only displayed this and the '**Mortality Rate by Age**' diagram.

Mortality Rate by Age (Line Chart)

What

Trend of mortality rate across different age groups.

Why

To examine how age impacts mortality risk.

How

Line plot showing mortality rate for each age category.

What we learned

Mortality risk increases with age, especially in older patient groups.

Top Predictors of Mortality (Feature Importance)

What

Identification of the most important features influencing mortality prediction.

Why

To understand which clinical variables contribute most to predicting high-risk patients..

How

Feature importance scores were extracted from the trained model to rank predictors.

What we learned

Key predictors include ventilation, vasopressor use, sepsis, and severity scores such as APACHE and SOFA, showing strong influence on mortality risk.

Exploratory Data Analysis (EDA)

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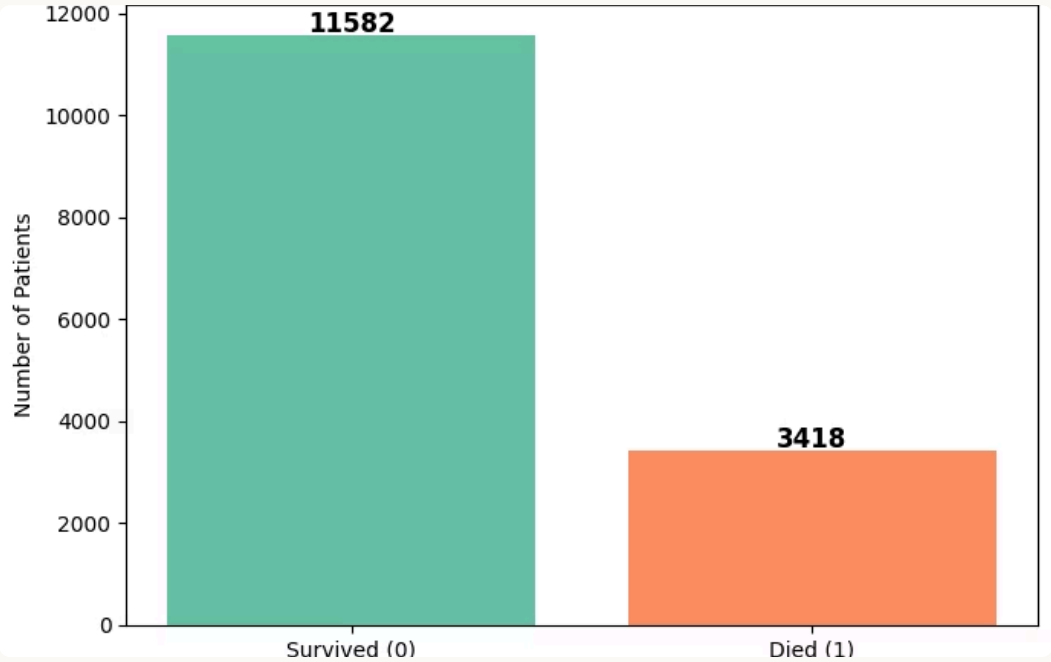
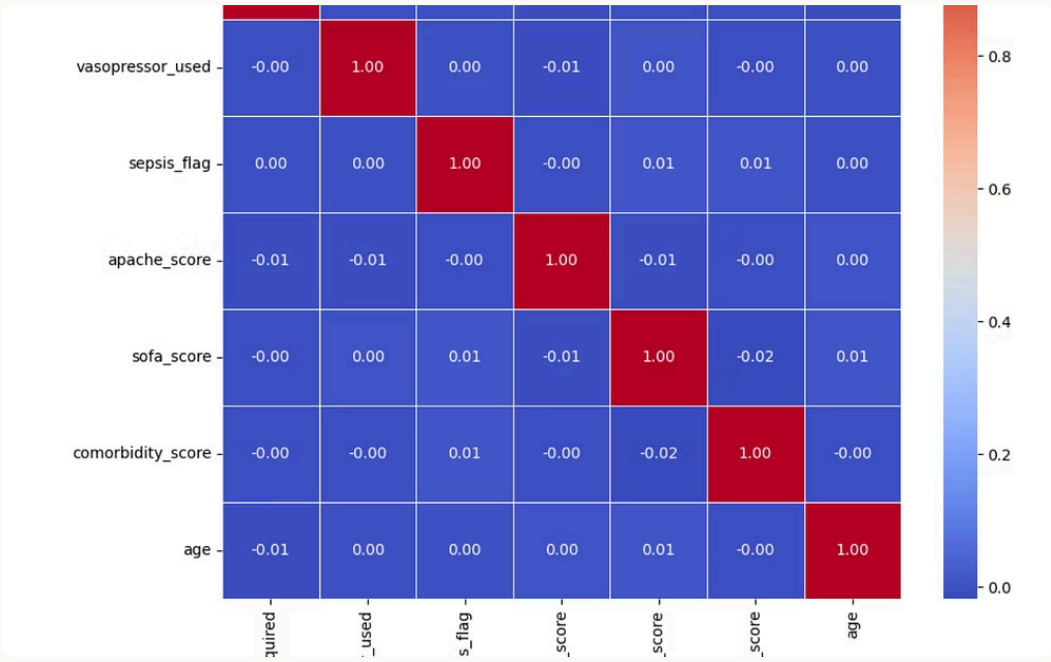




Class distribution (Bar chart)

What:
Shows the distribution of survived vs. died patients in the dataset.

Why:
To highlight the **class imbalance** problem.

When:
During the **EDA phase before model training**.

How:
By counting each class and visualizing it using a **bar chart**.

What we learned

The dataset is imbalanced, so special handling is required to avoid biased predictions.

Correlation Heatmap

What:
Shows the relationships between the selected clinical features.

Why:
To identify correlations and ensure features are not redundant.

When:
During the EDA phase before model training.

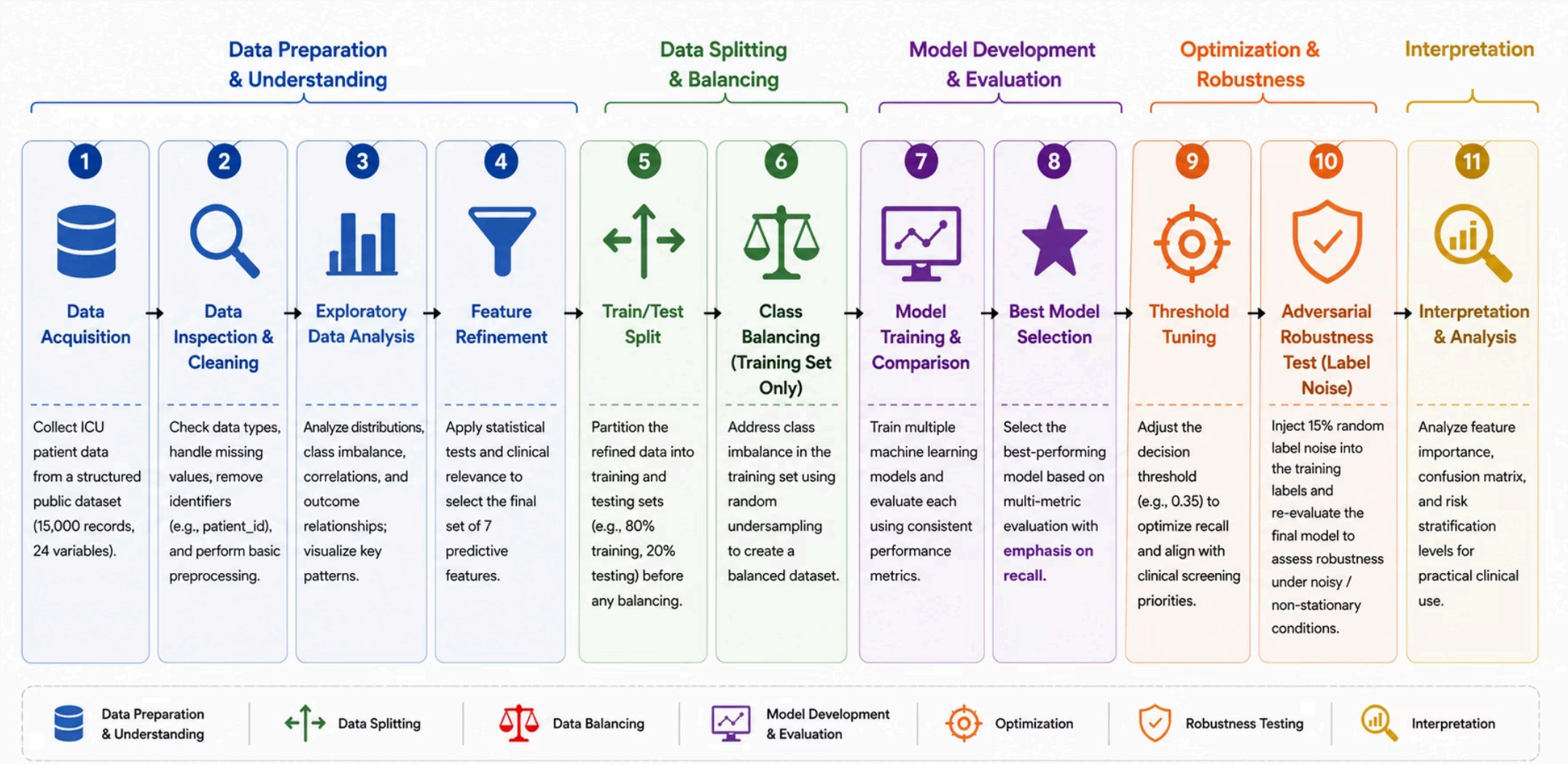
How:
By calculating pairwise correlations and visualizing them using a heatmap.

What we learned:
The features show low correlation, meaning minimal redundancy, while severity scores capture different aspects of patient condition, making the feature set suitable for modeling.

Key Insight: Several features showed statistically significant differences between survivors and non-survivors, confirming their predictive potential for mortality classification.

Critical Finding: Key predictors include ventilation requirement, vasopressor use, sepsis status, age, comordability score, and severity scores (APACHE, SOFA).

Methodology Workflow (Experiment 3)



Experiments Overview

Three progressive experiments were designed to iteratively improve model performance — moving from a broad baseline to a refined, clinically optimized final model.

Experiment 1 — Baseline Model

Trained a standard Logistic Regression on all available features with minimal preprocessing. Establishes a performance floor and reveals initial challenges with class imbalance.

Experiment 2 — Feature Selection

Applied importance-based feature selection to retain only the most predictive variables. Reduced noise and improved model generalizability while decreasing computational overhead.

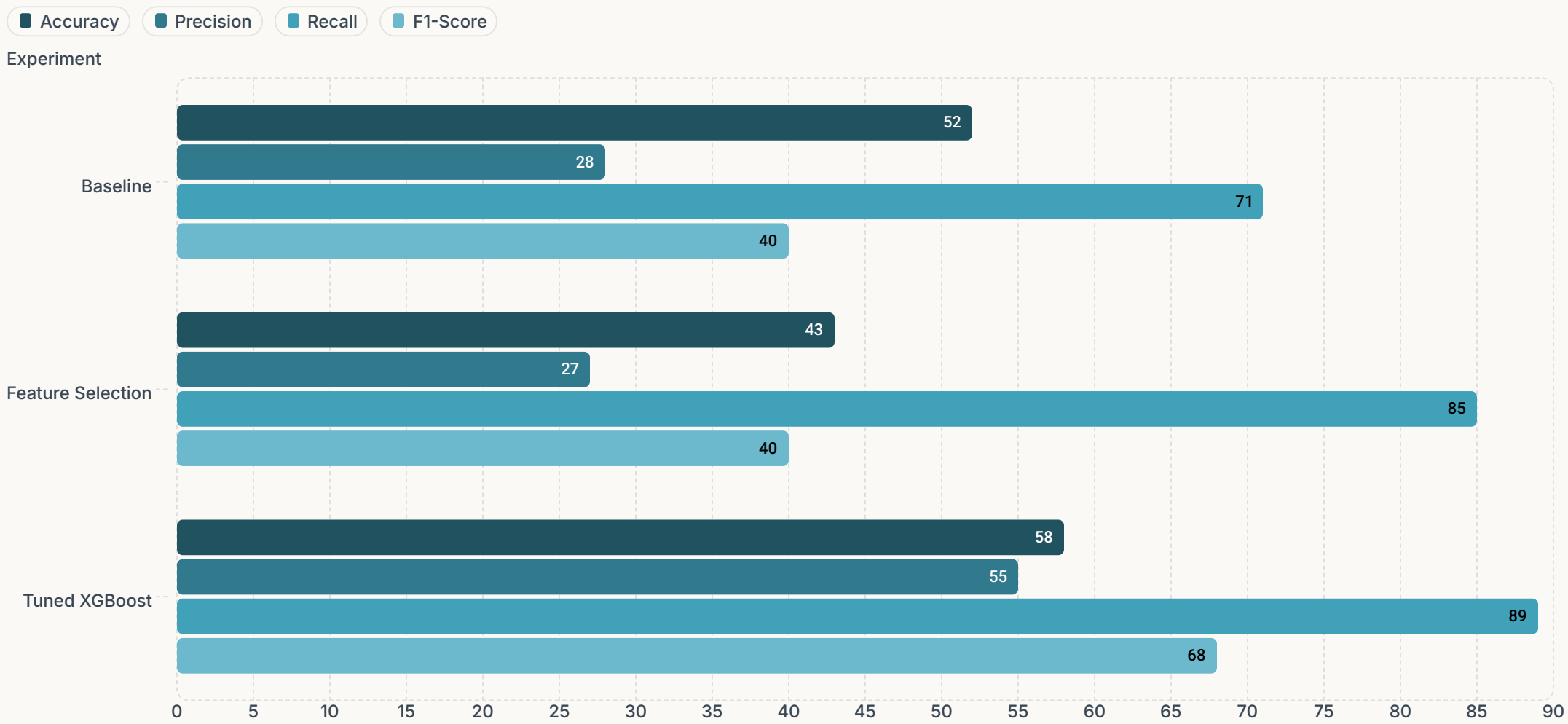
Experiment 3 — Final Optimized Model

Incorporated noise handling, class balancing (e.g., SMOTE or class weights), and hyperparameter tuning to maximize Recall for the high-risk class without sacrificing overall performance.

Experiment Results — Performance Comparison

Each experiment was evaluated using four standard classification metrics. The progressive improvement across experiments validates the iterative design approach, with **Recall for Class 1 (mortality)** being the most clinically critical measure.

Experiment	Accuracy	Precision	Recall	F1-Score
Exp 1 — Baseline (Logistic Regression, all features)	0.523	0.284	0.715	0.406
Exp 2 — Feature Selection (top predictors only)	0.428	0.265	0.851	0.404
Exp 3 — Tuned XGBoost (balanced + tuned)	**0.580**	**0.550**	**0.885**	**0.678**



> Final performance achieved using tuned XGBoost with class balancing and threshold optimization (0.35).

Final Model & Evaluation

The **Experiment 3 optimized model** was selected as the final solution based on its superior performance across all evaluation metrics — particularly its dramatically improved Recall for the mortality class.

In healthcare, **Recall (Sensitivity)** is the most critical metric: a missed high-risk patient (false negative) can have irreversible consequences, whereas a false positive typically results in additional — not harmful — clinical review.

Final Recall: 0.88 — meaning the model correctly identifies 88% of patients who will not survive, enabling timely clinical intervention.

58%

Accuracy

Overall correct classifications across both classes.

88%

Recall

High-risk patient detection rate — the primary clinical metric.

67%

F1-Score

Balanced precision-recall tradeoff for the mortality class.

Solution & Prototype

The final model functions as a **clinical decision-support tool** — it ingests structured patient data and outputs both a risk probability and a binary classification, giving clinicians an interpretable signal to act on. The model prioritizes recall to minimize missed high-risk patients, which is critical in ICU settings.

1

Data Input

Patient clinical features (e.g., age, ventilation, severity scores) are entered into the trained model.

2

Risk Probability Output

The model outputs a probability score (0–1) representing mortality risk.

3

Binary Classification

A threshold is applied to classify patients as: **Low Risk (0) and High Risk (1)**

Example Model Output

Field	Value
Risk Probability	0.47 (46.56%)
Threshold	0.35
Classification	High Risk (1)
Risk Category	Medium Risk

A lower threshold (0.35) was selected to improve **recall**, ensuring high-risk patients are not missed.

Google Docs

Screen Recording 1447-11-09 at 4.12.21 PM.mov

Conclusion

Iterative Experimentation

Three structured experiments were conducted, including baseline modeling, feature refinement, and model optimization. The final pipeline integrates feature selection, class balancing, and threshold tuning to improve clinical relevance.

Measurable Improvement

Model performance improved across experiments, with recall increasing from **~0.71 to 0.885**, significantly enhancing high-risk patient detection. The final model maintains a strong balance between recall, F1-score, and ROC-AUC.

Reliable Predictions

The final model identifies key predictors including ventilation requirement, vasopressor use, sepsis status, APACHE score, SOFA score, comorbidity score, and age. These features provide clinically interpretable and consistent mortality risk estimation.

> The proposed framework delivers a **recall-focused, interpretable, and reproducible solution** for ICU mortality risk prediction, supporting early detection and clinical decision-making.
>
> *(Cabitza et al., 2020)*

Future Work

While the current model delivers strong results, several directions can further enhance its performance, reliability, and real-world applicability in clinical environments.

1

Improve Dataset & Generalizability

Extend the model using multi-center ICU datasets with diverse patient populations to improve generalization and real-world applicability.

2

Advanced Modeling & Optimization

Explore cost-sensitive learning, SMOTE-based balancing, and deeper hyperparameter tuning to further improve mortality-case detection and model robustness.

3

Temporal Data Integration

Incorporate time-series clinical data such as vital signs and lab trends to better capture patient deterioration over time.

4

Model Interpretability & Calibration

Enhance interpretability using explainability methods (e.g., SHAP) and perform probability calibration to improve clinical trust and usability.

5

Real-World Deployment & Validation

Validate the model through prospective clinical testing, including clinician oversight, regulatory review, and integration into hospital decision-support systems.

Future work focuses on improving **generalizability, robustness, and clinical applicability** to enable safe and effective real-world deployment.

Reference

Kim et al. (2019). Machine learning for ICU mortality prediction

Li et al. (2021). ICU mortality prediction using MIMIC-IV data

Zhang et al. (2020). Mortality prediction using SOFA score and age

Cabitza et al. (2020). Challenges of AI in healthcare

Chen T & Guestrin C (2016). XGBoost: A scalable tree boosting system

Pedregosa F et al. (2011). Scikit-learn: Machine Learning in Python

Kaggle ICU Datasets: **https://www.kaggle.com/datasets/jayjoshi37/icu-patient-monitoring-and-mortality-prediction**

Thank You

Any Questions?

Project: Mortality Risk Prediction using Machine Learning

Course: Data Science in healthcare

We welcome your feedback and questions